Mathematical Formulation of the Software Team Formation Problem as a Mixed-Integer Linear Program

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1 Problem Definition

This work investigates the Software Team Formation Problem (STFP). An instance of the problem is defined by a target project P with technical requirements specified by dimension and priority, a fixed team size k, a set of available developers

$$D = \{d_1, d_2, \ldots, d_N\},$$

and a collaboration graph $G = (D, E)$.

In G, the vertices represent developers, and an edge $(d_i, d_j) \in E$ indicates that developers d_i and d_j have previously collaborated, as observed in the history of participation in projects. This edge is weighted by w_{ij}, which represents the intensity of prior collaboration as a function of the number of projects shared by the two developers.

The objective of the STFP in this work is to select the team with the greatest fitness to execute the target project.

A team's fitness is determined by two complementary components:

- **Technical Adequacy (AT):** the correspondence between the technical competencies of the selected developers and the technical requirements of the target project. These requirements are organized into three dimensions: application domain (DA), development ecosystem (ED), and programming language (LP). Within each dimension, requirements are prioritized using the MoSCoW method, which classifies them as Must-have, Should-have, Could-have, and Won't-have. Because Won't-have requirements are outside the scope of the current project version, only the first three categories are considered in the formulation.
- **Collaborative Adequacy (AC):** the team's potential collaborative quality, estimated from pairwise compatibility between developers based on their history of working together on previous projects.

These two components are integrated into an overall fitness, Team Adequacy (AE), which represents the final criterion for evaluating a team with respect to the target project.

A candidate team is a subset of developers. A team is feasible when it contains exactly k developers and covers every Must-have requirement defined for the target project. Thus, Must-have requirements are treated as hard constraints of the model rather than merely as penalties in the objective function. Among all feasible teams, the goal is to find the one that maximizes

overall fitness with respect to the target project:

$$\max_{\substack{T \subseteq D,\ |T|=k,\\ T \text{ covers all Must requirements}}} AE(T, P),$$

where D is the set of available developers, T is a candidate team, P is the target project, k is the fixed team size, and $AE(T, P)$ is the team's overall fitness.

The formulation is built directly as a Mixed-Integer Linear Programming (MILP) model that combines binary and continuous variables. Nonlinearities in the surrogate evaluation model are handled inside the model itself: products between binary variables are linearized through exact McCormick constraints, the lower and upper branches of the clamp operator are represented by bounded variables and, when required, a binary indicator, the penalty associated with Must-have requirements is modeled through discrete levels, and the global mixminmax aggregation is handled by decomposition into two linear subproblems. The resulting model can be solved directly by MILP solvers.

2 Mathematical Formulation

2.1 Problem Inputs

Let

$$D = \{d_1, d_2, \ldots, d_N\}$$

be the set of available developers, filtered to include only developers present in the collaboration graph. Each developer $d_i \in D$ has technical competencies associated with three dimensions:

$$\Delta = \{\text{DA}, \text{ED}, \text{LP}\},$$

where DA denotes application domain, ED denotes development ecosystem, and LP denotes programming language.

For each dimension $\delta \in \Delta$, let U^δ be the universe of possible competencies in that dimension. Developer d_i's competencies in dimension δ are represented by the binary parameter

$$a_{iu}^\delta = \begin{cases} 1, & \text{if } d_i \text{ has competency } u \text{ in dimension } \delta, \\ 0, & \text{otherwise,} \end{cases}$$

for every $d_i \in D$, $u \in U^\delta$, and $\delta \in \Delta$.

Additionally, for each pair of developers (d_i, d_j) with a recorded collaboration history, define the raw collaborative compatibility $PC_{ij} \in [0, 1]$, estimated from the actual collaboration history between d_i and d_j. By construction, $PC_{ij} = PC_{ji}$. The set E contains the pairs with recorded collaboration. To reproduce the behavior of the surrogate evaluator used by the GA, each PC_{ij} is converted into a discretized score $\widehat{PC}_{ij}$, associated with the VL, L, M, H, and VH bands, with centroids 0.1, 0.3, 0.5, 0.7, and 0.9. For pairs outside E, that is, with no prior collaboration record, the constant value

$$\widehat{PC}_{\text{low}} = 0.3$$

is adopted, corresponding to the Low band, based on the rule elicited with the expert.

The target project is represented by P. For each technical dimension $\delta \in \Delta$, the project defines requirements organized by MoSCoW priority:

$$R_P^\delta = \left(M_P^\delta, S_P^\delta, C_P^\delta\right),$$

where $M_P^\delta \subseteq U^\delta$ is the set of Must-have requirements, $S_P^\delta \subseteq U^\delta$ is the set of Should-have requirements, and $C_P^\delta \subseteq U^\delta$ is the set of Could-have requirements; these sets are disjoint by construction.

The regression coefficients

$$(b_1, b_2, w_{\text{covM}}, w_{\text{red2M}}, w_{\text{red3M}}, w_{\text{red4M}}, w_{\text{should}}, w_{\text{could}}, a_{\text{full}}, a_{\text{red2}})$$

are model constants. The dimension weights $(w_{\text{DA}}, w_{\text{ED}}, w_{\text{LP}})$, the global aggregation weights $(w_{\min}, w_{\max})$, the floor parameter $\eta \in [0, 1]$, and the neutral value $\mu_{\text{neutral}} \in [0, 1]$ are model constants. The collaborative centroids $\{0.1, 0.3, 0.5, 0.7, 0.9\}$, the no-collaboration value $\widehat{PC}_{\text{low}} = 0.3$, and the collaborative compression factor $\gamma = 0.5$ are also model constants. The coefficients and weights were previously calibrated against the Bayesian Network; the compression of AC and the independent decomposition of the Should-have and Could-have contributions correspond to the refinement conducted after validation with the decision authority.

2.2 Decision Variables

The primary decision variable represents team selection:

$$x_i = \begin{cases} 1, & \text{if developer } d_i \text{ is selected for the team,} \\ 0, & \text{otherwise,} \end{cases} \qquad \forall d_i \in D,$$

subject to the team-size constraint

$$\sum_{i=1}^{N} x_i = k.$$

In addition to x_i, the formulation introduces auxiliary variables for coverage, products, penalties, and aggregation as each nonlinearity is linearized. They are consolidated in Section 2.9.

2.3 Competency Counts and Coverage

For each competency $u \in U^\delta$, define the number of team members who possess it:

$$h_u^\delta(x) = \sum_{i=1}^{N} a_{iu}^\delta x_i, \qquad \forall \delta \in \Delta,\ \forall u \in U^\delta.$$

This expression is linear: it is a sum of constants a_{iu}^δ multiplied by variables x_i.

Simple coverage, meaning that at least one member has the competency, is represented by the binary variable y_u^δ. For each level $r \in \{2, 3, 4\}$, redundant coverage by at least r members is represented by the binary variable $z_{r,u}^\delta$. These variables are defined through the linear constraints

$$y_u^\delta \le h_u^\delta(x), \qquad y_u^\delta \ge \frac{1}{k} h_u^\delta(x), \qquad \forall \delta \in \Delta,\ \forall u \in U^\delta,$$

$$r z_{r,u}^\delta \le h_u^\delta(x), \qquad z_{r,u}^\delta \ge \frac{1}{k}\left(h_u^\delta(x) - (r-1)\right),$$

$$\forall \delta \in \Delta,\ \forall u \in U^\delta,\ \forall r \in \{2, 3, 4\},$$

$$y_u^\delta, z_{r,u}^\delta \in \{0, 1\}, \qquad \forall \delta \in \Delta,\ \forall u \in U^\delta,\ \forall r \in \{2, 3, 4\}.$$

Must-have requirements are also imposed as hard feasibility constraints. For each mandatory competency $u \in M_P^\delta$, at least one selected developer must possess that competency:

$$h_u^\delta(x) \ge 1, \qquad \forall \delta \in \Delta,\ \forall u \in M_P^\delta.$$

This constraint ensures that a team is feasible only if it fully covers all mandatory requirements of the target project.

2.4 Coverage Features by MoSCoW Priority

For each dimension $\delta \in \Delta$ and priority $\pi \in \{M, S, C\}$, let $R_P^{\delta,\pi}$ be the set of project requirements at that priority.

Simple coverage (cov).

$$\mathrm{cov}^{\delta,\pi} = \begin{cases} \dfrac{1}{|R_P^{\delta,\pi}|} \displaystyle\sum_{u \in R_P^{\delta,\pi}} y_u^{\delta}, & \text{if } |R_P^{\delta,\pi}| > 0, \\ 1, & \text{if } |R_P^{\delta,\pi}| = 0. \end{cases}$$

Redundant coverage (red2, red3, and red4).

$$\mathrm{red}_r^{\delta,\pi} = \begin{cases} \dfrac{1}{|R_P^{\delta,\pi}|} \displaystyle\sum_{u \in R_P^{\delta,\pi}} z_{r,u}^{\delta}, & \text{if } |R_P^{\delta,\pi}| > 0, \\ 0, & \text{if } |R_P^{\delta,\pi}| = 0, \end{cases} \qquad r \in \{2, 3, 4\}.$$

For notational simplicity, write

$$\mathrm{cov}_M^{\delta} = \mathrm{cov}^{\delta,M}, \quad \mathrm{cov}_S^{\delta} = \mathrm{cov}^{\delta,S}, \quad \mathrm{cov}_C^{\delta} = \mathrm{cov}^{\delta,C},$$

and analogously for redundancies. In the cross-priority terms of Case C, an absent Should-have or Could-have priority contributes zero, whereas an absent Must-have priority uses the neutral value $\mathrm{cov}_M^{\delta} = 1$. Every feature belongs to $[0, 1]$ and is a linear combination of binary variables divided by a constant.

2.5 Technical Score by Dimension (Linearized)

The technical score DIM^{δ} of each dimension depends on the set of active priorities:

$$\Pi^{\delta}(P) = \left\{ \pi \in \{M, S, C\} \mid |R_P^{\delta,\pi}| > 0 \right\}.$$

Each case below is already presented in its linear form, with auxiliary variables introduced where they replace a nonlinear operation.

Case A - no active priority ($|\Pi^{\delta}(P)| = 0$). The dimension is neutral and contributes no nonlinear terms: its score is the constant μ_{neutral} (see Section 2.6).

Case B - exactly one active priority ($|\Pi^{\delta}(P)| = 1$). Let π^* be the active priority. Conceptually, the score is $\mathrm{clamp}_{[0,1]}(v_B^{\delta})$, with linear argument

$$v_B^{\delta} = b_2 + a_{\text{full}} \, \mathrm{cov}^{\delta,\pi^*} + a_{\text{red2}} \, \mathrm{red}_2^{\delta,\pi^*}.$$

The single-priority coefficients used in both evaluated versions are

$$b_2 = -\frac{1}{3}, \qquad a_{\text{full}} = \frac{1}{6}, \qquad a_{\text{red2}} = \frac{7}{6}.$$

Consequently, $v_B^{\delta} \in [L_B, U_B]$, with $L_B = -1/3$ and $U_B = 1$. The operator

$$\mathrm{clamp}_{[0,1]}(v) = \max(0, \min(1, v))$$

is nonlinear. Because v_B^{δ} can be negative, an objective-driven upper bound alone is insufficient. Introduce $q_B^{\delta} \in \{0, 1\}$ and $\mathrm{DIM}_B^{\delta} \in [0, 1]$, with the exact linear constraints

$$\begin{aligned} &\mathrm{DIM}_B^{\delta} \geq v_B^{\delta}, \qquad && \mathrm{DIM}_B^{\delta} \leq v_B^{\delta} - L_B(1 - q_B^{\delta}), \\ &\mathrm{DIM}_B^{\delta} \leq U_B q_B^{\delta}, && 0 \leq \mathrm{DIM}_B^{\delta} \leq 1. \end{aligned}$$

If $v_B^\delta < 0$, these constraints force $q_B^\delta = 0$ and $\mathrm{DIM}_B^\delta = 0$; otherwise they force $q_B^\delta = 1$ and $\mathrm{DIM}_B^\delta = v_B^\delta$. Since $U_B = 1$, this is exactly $\mathrm{clamp}_{[0,1]}(v_B^\delta)$. The dimension score is

$$\mathrm{DIM}^\delta = \mathrm{DIM}_B^\delta.$$

Case C - two or more active priorities ($|\Pi^\delta(P)| \geq 2$). Conceptually,

$$\begin{aligned}\mathrm{DIM}^\delta = \mathrm{clamp}_{[0,1]} \Big(& b_1 + w_{\mathrm{covM}}\,\mathrm{cov}_M^\delta + w_{\mathrm{red2M}}\,\mathrm{red}_{2,M}^\delta \\ & + w_{\mathrm{red3M}}\,\mathrm{red}_{3,M}^\delta + w_{\mathrm{red4M}}\,\mathrm{red}_{4,M}^\delta + w_{\mathrm{should}}\,\underbrace{\mathrm{cov}_M^\delta \mathrm{cov}_S^\delta}_{\text{product}} \\ & + w_{\mathrm{could}}\,\underbrace{\mathrm{cov}_M^\delta \mathrm{cov}_C^\delta}_{\text{product}} \Big)\ \underbrace{\rho^\delta}_{\max(\eta,\cdot)}\ .\end{aligned}$$

The contributions of Should-have and Could-have are independent: increasing Should-have coverage does not reduce the Could-have contribution. When the dimension has no Must-have requirements, $\mathrm{cov}_M^\delta = 1$ is used as the neutral value. The coefficients used in the refined model are

$$b_1 = 0, \qquad w_{\mathrm{covM}} = 0, \qquad w_{\mathrm{red2M}} = 0.4120879121, \qquad w_{\mathrm{red3M}} = 0.2582417582,$$
$$w_{\mathrm{red4M}} = 0, \qquad w_{\mathrm{should}} = w_{\mathrm{could}} = 0.2307692308.$$

The term $\mathrm{red}_{4,M}^\delta$ is retained for traceability, but its weight is zero because the additional benefit of fourth-person coverage was not supported by elicitation. The two coverage products and the remaining nonlinear operations are handled next.

(C.1) Product $\mathrm{cov}_M^\delta \mathrm{cov}_S^\delta$. Because each $\mathrm{cov}^{\delta,\pi}$ is a sum of binary variables divided by a constant, the product is expanded before division. For each pair (i,j) such that $u_i \in R_P^{\delta,M}$ and $u_j \in R_P^{\delta,S}$, introduce $w_{ij}^{MS} \in \{0,1\}$ with exact McCormick constraints:

$$w_{ij}^{MS} \leq y_{u_i}^\delta, \qquad w_{ij}^{MS} \leq y_{u_j}^\delta, \qquad w_{ij}^{MS} \geq y_{u_i}^\delta + y_{u_j}^\delta - 1.$$

The linearized product is

$$p^\delta = \frac{1}{|R_P^{\delta,M}|\,|R_P^{\delta,S}|} \sum_i \sum_j w_{ij}^{MS}.$$

(C.2) Product $\mathrm{cov}_M^\delta \mathrm{cov}_C^\delta$. For each pair (i,j) such that $u_i \in R_P^{\delta,M}$ and $u_j \in R_P^{\delta,C}$, introduce $w_{ij}^{MC} \in \{0,1\}$:

$$w_{ij}^{MC} \leq y_{u_i}^\delta, \qquad w_{ij}^{MC} \leq y_{u_j}^\delta, \qquad w_{ij}^{MC} \geq y_{u_i}^\delta + y_{u_j}^\delta - 1.$$

The linearized product is

$$q^\delta = \frac{1}{|R_P^{\delta,M}|\,|R_P^{\delta,C}|} \sum_i \sum_j w_{ij}^{MC}.$$

When there are no Must-have requirements, the implementations directly use

$$p^\delta = \mathrm{cov}_S^\delta \qquad \text{and} \qquad q^\delta = \mathrm{cov}_C^\delta.$$

(C.3) Clamp of the intermediate score. With p^δ and q^δ already linear, the argument of the Case C score is

$$\begin{aligned} v_C^\delta = {} & b_1 + w_{\mathrm{covM}}\mathrm{cov}_M^\delta + w_{\mathrm{red2M}}\mathrm{red}_{2,M}^\delta + w_{\mathrm{red3M}}\mathrm{red}_{3,M}^\delta \\ & + w_{\mathrm{red4M}}\mathrm{red}_{4,M}^\delta + w_{\mathrm{should}}p^\delta + w_{\mathrm{could}}q^\delta. \end{aligned}$$

The clamp is linearized through the variable $\mathrm{DIM}_{\mathrm{clamp}}^\delta \in [0,1]$:

$$\mathrm{DIM}_{\mathrm{clamp}}^\delta \leq v_C^\delta, \qquad 0 \leq \mathrm{DIM}_{\mathrm{clamp}}^\delta \leq 1.$$

(C.4) Must penalty through discrete levels. In the code used in the experiments, the penalty associated with Must-have requirements is not modeled as a product of two continuous variables. Instead, an exact discrete representation is used. Let

$$n_M^\delta = |M_P^\delta|$$

be the number of mandatory requirements in the dimension, and let

$$K_M^\delta = \sum_{u \in M_P^\delta} y_u^\delta$$

be the number of mandatory requirements covered.

For each level $q \in \{0, 1, \ldots, n_M^\delta\}$, define the constant

$$\rho_q^\delta = \max\left(\eta, \frac{q}{n_M^\delta}\right).$$

Introduce the binary variable $r_q^\delta \in \{0, 1\}$, which indicates the active mandatory-coverage level:

$$\sum_{q=0}^{n_M^\delta} r_q^\delta = 1, \qquad K_M^\delta = \sum_{q=0}^{n_M^\delta} q\, r_q^\delta.$$

Next, linearize the product between $\mathrm{DIM}_{\mathrm{clamp}}^\delta$ and the activated binary level. Introduce $s_q^\delta \in [0, 1]$ such that

$$s_q^\delta \le \mathrm{DIM}_{\mathrm{clamp}}^\delta, \qquad s_q^\delta \le r_q^\delta,$$
$$s_q^\delta \ge \mathrm{DIM}_{\mathrm{clamp}}^\delta - (1 - r_q^\delta), \qquad s_q^\delta \ge 0.$$

The final Case C dimension score is

$$t^\delta = \sum_{q=0}^{n_M^\delta} \rho_q^\delta s_q^\delta.$$

When $n_M^\delta = 0$, there is no Must penalty, and

$$t^\delta = \mathrm{DIM}_{\mathrm{clamp}}^\delta.$$

Thus, the only product linearized at this stage is between a continuous variable and a binary variable, preserving MILP exactness. The Case C dimension score is

$$\mathrm{DIM}^\delta = t^\delta.$$

Version-specific parameterizations. For multiple-priority dimensions, the original calibrated evaluator (V1) uses the pre-penalty argument

$$v_{C,\mathrm{V1}}^\delta = 0.071429\, \mathrm{cov}_M^\delta + 0.428571\, \mathrm{red}_{2,M}^\delta + 0.357143\, \mathrm{cov}_M^\delta \mathrm{cov}_S^\delta + 0.5\, \mathrm{cov}_M^\delta \mathrm{cov}_C^\delta \left(1 - \mathrm{cov}_S^\delta\right).$$

The refined evaluator (V2), described in detail above, uses

$$v_{C,\mathrm{V2}}^\delta = 0.4120879121\, \mathrm{red}_{2,M}^\delta + 0.2582417582\, \mathrm{red}_{3,M}^\delta + 0.2307692308\, \mathrm{cov}_M^\delta \mathrm{cov}_S^\delta + 0.2307692308\, \mathrm{cov}_M^\delta \mathrm{cov}_C^\delta.$$

Both versions use $w_{\mathrm{DA}} = 3$, $w_{\mathrm{ED}} = 1$, and $w_{\mathrm{LP}} = 5$, followed by the global mixminmax weights $w_{\min} = 5$ and $w_{\max} = 1$. V1 uses the original collaboration score AC_0; V2 uses the compressed score $AC_c = 0.5 + 0.5(AC_0 - 0.5)$. The V1 and V2 Case C arguments are nonnegative over their feasible feature domains, so their lower clamp branch is inactive; the upper bound $\mathrm{DIM}_{\mathrm{clamp}}^\delta \le 1$, together with maximization, implements the remaining upper clamp. The products in the V1 expression are expanded with the same exact binary-product construction used above, with one additional product layer for the three-factor term.

2.6 Team Technical Adequacy

Technical adequacy is the weighted aggregation of the linearized dimension scores, with neutral handling:

$$\widehat{\mathrm{DIM}}^{\delta} = \begin{cases} \mathrm{DIM}^{\delta}, & \text{if } |\Pi^{\delta}(P)| \geq 1 \quad \text{(Case B or C)}, \\ \mu_{\text{neutral}}, & \text{if } |\Pi^{\delta}(P)| = 0 \quad \text{(Case A)}. \end{cases}$$

Then

$$AT(x,P) = \frac{w_{\mathrm{DA}}\widehat{\mathrm{DIM}}^{\mathrm{DA}} + w_{\mathrm{ED}}\widehat{\mathrm{DIM}}^{\mathrm{ED}} + w_{\mathrm{LP}}\widehat{\mathrm{DIM}}^{\mathrm{LP}}}{w_{\mathrm{DA}} + w_{\mathrm{ED}} + w_{\mathrm{LP}}}.$$

Because $w_{\mathrm{DA}}, w_{\mathrm{ED}}, w_{\mathrm{LP}}$ are constants and each $\widehat{\mathrm{DIM}}^{\delta}$ is an already linearized variable or constant, AT is linear. The weights reproduce the calibrated wmean function at the Bayesian Network's AT node:

$$w_{\mathrm{DA}} = 3, \qquad w_{\mathrm{ED}} = 1, \qquad w_{\mathrm{LP}} = 5.$$

2.7 Team Collaborative Adequacy

Collaborative adequacy is calculated over the

$$\binom{k}{2}$$

pairs of developers in the selected team. To maintain equivalence with the surrogate evaluator used by the GA, the ILP does not directly use the raw continuous value PC_{ij}. Before aggregation, each PC_{ij} is discretized into collaborative bands with the same centroids used by the surrogate:

$$\widehat{PC}_{ij} = \begin{cases} 0.1, & \text{if } PC_{ij} < 0.2, \\ 0.3, & \text{if } 0.2 \leq PC_{ij} < 0.4, \\ 0.5, & \text{if } 0.4 \leq PC_{ij} < 0.6, \\ 0.7, & \text{if } 0.6 \leq PC_{ij} < 0.8, \\ 0.9, & \text{if } PC_{ij} \geq 0.8. \end{cases}$$

For each pair (d_i, d_j) with a recorded edge in the collaboration graph, introduce the binary variable p_{ij}, which indicates whether both developers were selected:

$$p_{ij} \leq x_i, \qquad p_{ij} \leq x_j, \qquad p_{ij} \geq x_i + x_j - 1, \qquad \forall (i,j) \in E.$$

Because $x_i, x_j \in \{0,1\}$, this linearization is exact.

Team pairs with no prior collaboration record in the graph receive the fixed weight

$$\widehat{PC}_{\text{low}} = 0.3,$$

corresponding to the Low band. The original collaborative adequacy is

$$AC_0(x) = \frac{1}{\binom{k}{2}} \left[\sum_{(i,j)\in E} \widehat{PC}_{ij} p_{ij} + \widehat{PC}_{\text{low}} \left(\binom{k}{2} - \sum_{(i,j)\in E} p_{ij} \right) \right].$$

In the refined model, this score is compressed around the neutral point 0.5, with $\gamma = 0.5$:

$$AC_c(x) = 0.5 + \gamma \left(AC_0(x) - 0.5 \right) = 0.25 + 0.5 AC_0(x).$$

This transformation preserves the center and the ordering of teams but halves their distances from 0.5. Because $\widehat{PC}_{ij}$, $\widehat{PC}_{\text{low}}$, $\binom{k}{2}$, and γ are constants, and p_{ij} is a linearized binary variable, both $AC_0(x)$ and $AC_c(x)$ are linear expressions.

2.8 Overall Team Adequacy (mixminmax by Decomposition)

Overall adequacy combines AT and compressed collaborative adequacy AC_c through the mixminmax function, with calibrated bottleneck effect $w_{\min} = 5$ and $w_{\max} = 1$:

$$AE(x, P) = \frac{w_{\min} \min(AT(x, P), AC_c(x)) + w_{\max} \max(AT(x, P), AC_c(x))}{w_{\min} + w_{\max}}.$$

In the experiments, this function is not solved through the earlier formulation with variables m, M, and a binary ordering variable. Instead, the problem is decomposed into two linear subproblems, one for each possible ordering between AT and AC_c.

Case A: $AT \leq AC_c$**.** In this case,

$$\min(AT, AC_c) = AT \qquad \text{and} \qquad \max(AT, AC_c) = AC_c.$$

The subproblem adds the constraint

$$AT(x, P) \leq AC_c(x)$$

and maximizes

$$AE_A(x, P) = \frac{w_{\min} AT(x, P) + w_{\max} AC_c(x)}{w_{\min} + w_{\max}}.$$

Case B: $AC_c \leq AT$**.** In this case,

$$\min(AT, AC_c) = AC_c \qquad \text{and} \qquad \max(AT, AC_c) = AT.$$

The subproblem adds the constraint

$$AC_c(x) \leq AT(x, P)$$

and maximizes

$$AE_B(x, P) = \frac{w_{\max} AT(x, P) + w_{\min} AC_c(x)}{w_{\min} + w_{\max}}.$$

The final value returned by the ILP is the better value from the two subproblems:

$$AE^\star(P) = \max\left(AE_A^\star(P), AE_B^\star(P)\right).$$

Because $AT(x, P)$ and $AC_c(x)$ are linear expressions, each subproblem is a linear MILP. The decomposition is exact because the two cases cover every possible ordering between AT and AC_c.

2.9 Complete Optimization Model (MILP)

Consolidating the previous sections, the model used in the experiments is the `AE_FULL_EXACT_BINS` mode: the ILP uses exact pairwise AC_0, discretized into the same bands as the GA surrogate evaluator and followed by the same affine compression. The model is solved by decomposition into two linear subproblems.

The constraints common to the two subproblems are

$$\sum_{i=1}^{N} x_i = k,$$

$$h_u^\delta = \sum_{i=1}^{N} a_{iu}^\delta x_i, \qquad \forall \delta \in \Delta,\ \forall u \in U^\delta,$$

$$y_u^\delta \le h_u^\delta, \qquad \forall \delta \in \Delta,\ \forall u \in U^\delta,$$

$$y_u^\delta \ge \frac{1}{k} h_u^\delta, \qquad \forall \delta \in \Delta,\ \forall u \in U^\delta,$$

$$r z_{r,u}^\delta \le h_u^\delta, \qquad \forall \delta \in \Delta,\ \forall u \in U^\delta,\ \forall r \in \{2,3,4\},$$

$$z_{r,u}^\delta \ge \frac{1}{k}\left(h_u^\delta - (r-1)\right), \qquad \forall \delta \in \Delta,\ \forall u \in U^\delta,\ \forall r \in \{2,3,4\},$$

$$h_u^\delta \ge 1, \qquad \forall \delta \in \Delta,\ \forall u \in M_P^\delta.$$

The dimension-level technical scores are calculated according to Cases A, B, and C in Section 2.5. For dimensions in Case B,

$$\mathrm{DIM}_B^\delta \ge v_B^\delta, \qquad \mathrm{DIM}_B^\delta \le v_B^\delta - L_B(1 - q_B^\delta),$$

$$\mathrm{DIM}_B^\delta \le U_B q_B^\delta, \qquad 0 \le \mathrm{DIM}_B^\delta \le 1, \qquad q_B^\delta \in \{0,1\}.$$

For dimensions in Case C, the exact linearizations of the products w^{MS} and w^{MC}, the clamp variable $\mathrm{DIM}_{\mathrm{clamp}}^\delta$, and the discrete-level Must penalty are used:

$$\sum_{q=0}^{n_M^\delta} r_q^\delta = 1, \qquad K_M^\delta = \sum_{q=0}^{n_M^\delta} q r_q^\delta,$$

$$s_q^\delta \le \mathrm{DIM}_{\mathrm{clamp}}^\delta, \qquad s_q^\delta \le r_q^\delta, \qquad s_q^\delta \ge \mathrm{DIM}_{\mathrm{clamp}}^\delta - (1 - r_q^\delta),$$

$$t^\delta = \sum_{q=0}^{n_M^\delta} \rho_q^\delta s_q^\delta, \qquad \rho_q^\delta = \max\left(\eta, \frac{q}{n_M^\delta}\right).$$

Technical adequacy is

$$AT(x,P) = \frac{w_{\mathrm{DA}} \widehat{\mathrm{DIM}}^{\mathrm{DA}} + w_{\mathrm{ED}} \widehat{\mathrm{DIM}}^{\mathrm{ED}} + w_{\mathrm{LP}} \widehat{\mathrm{DIM}}^{\mathrm{LP}}}{w_{\mathrm{DA}} + w_{\mathrm{ED}} + w_{\mathrm{LP}}},$$

with $w_{\mathrm{DA}} = 3$, $w_{\mathrm{ED}} = 1$, and $w_{\mathrm{LP}} = 5$.

The original and compressed collaborative adequacy scores are

$$AC_0(x) = \frac{1}{\binom{k}{2}} \left[\sum_{(i,j) \in E} \widehat{PC}_{ij} p_{ij} + \widehat{PC}_{\mathrm{low}} \left(\binom{k}{2} - \sum_{(i,j) \in E} p_{ij} \right) \right],$$

$$AC_c(x) = 0.5 + 0.5\left(AC_0(x) - 0.5\right),$$

subject to

$$p_{ij} \le x_i, \qquad p_{ij} \le x_j, \qquad p_{ij} \ge x_i + x_j - 1, \qquad \forall (i,j) \in E.$$

The Case A subproblem is

$$\max AE_A = \frac{w_{\min} AT(x,P) + w_{\max} AC_c(x)}{w_{\min} + w_{\max}} \tag{STFP-A}$$

subject to

$$AT(x,P) \le AC_c(x)$$

and the common model constraints.

The Case B subproblem is

$$\max AE_B = \frac{w_{\max} AT(x,P) + w_{\min} AC_c(x)}{w_{\min} + w_{\max}} \tag{STFP-B}$$

subject to

$$AC_c(x) \leq AT(x, P)$$

and the common model constraints.

The final solution is

$$AE^\star = \max\left(AE_A^\star, AE_B^\star\right).$$

The global status is considered optimal only when both subproblems are solved to optimality. Otherwise, when the solver returns an incumbent solution, the result is reported as a feasible solution with the corresponding MIP gap.

The model variables have the following domains:

$$x_i \in \{0,1\}, \qquad y_u^\delta, z_{r,u}^\delta \in \{0,1\}, \qquad r \in \{2,3,4\}, \qquad p_{ij} \in \{0,1\},$$

$$w_{ij}^{MS}, w_{ij}^{MC}, r_q^\delta, q_B^\delta \in \{0,1\},$$

$$\mathrm{DIM}_B^\delta, \mathrm{DIM}_{\mathrm{clamp}}^\delta, t^\delta, s_q^\delta, AT, AC_0, AC_c, AE_A, AE_B \in [0,1].$$

In the experiments, the model was solved with Gurobi through PuLP, using an instance-specific time limit, `gapRel=0.0`, `MIPFocus=1`, and `Threads=0`. When the complete `AE_FULL_EXACT_BINS` mode is used, the time limit is divided between the two decomposition subproblems.

2.10 Experimental Correspondence Check

The solver-selected teams were independently rescored with the corresponding direct evaluator. Across the twelve controlled RQ2 projects, the maximum absolute difference between the solver objective and the directly recomputed AE was 4.17×10^{-7} for V1 and 5.67×10^{-7} for V2. No benchmark dimension in either RQ2 version had exactly one active priority; therefore, correcting the general Case B clamp does not change the reported teams, scores, or conclusions.

For these controlled projects, the global solution was recorded as optimal only when both ordering subproblems were solved to optimality. The recorded MILP runtime is the sum of the two subproblem runtimes; separate per-subproblem times and statuses were not retained in the result CSV.